Statistics for Data Science - II
Week-3, 4 and 5 questions

1. The joint PMF of two discrete random variables X and Y is given in the following table:

$Y \backslash X$	-1	0	$f_Y(y)$
-1	$\frac{1}{12}$	$\frac{1}{\alpha}$	$\frac{1}{12}+1/\alpha$
0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}+1/3$
1	$\frac{1}{4}-\frac{1}{\alpha}$	$\frac{1}{6}$	$\frac{1}{4}-\frac{1}{\alpha}+\frac{1}{6}$
$f_X(x)$			

In the table, α is any positive real number. Find the value of α such that $\mathrm{Cov}(X, Y) = 0$. [3 marks]

Answer: 6

$$E[XY] - E[X]\cdot E[Y] = 0$$

$$\Rightarrow E[XY] = E[X]\cdot E[Y]$$

$$(-1)\cdot(-1)\cdot\frac{1}{12} + 0 + 0 + 0 + (-1)\cdot(1)\left(\frac{1}{4}-\frac{1}{\alpha}\right) + 0 = \quad -$$

$$E[X] =$$

$$E[Y] = (-1)\left(\frac{1}{12}+\frac{1}{\alpha}\right) + 0 + 1\cdot\left(\frac{1}{4}-\frac{1}{\alpha}+\frac{1}{6}\right) =$$

2. The probability mass function of a random variable X is given as

x	-3	6	9
$P(X = x)$	$1/6$	$1/2$	$1/3$

Define $Y = (2X + 1)^2$. Find the expected value of Y. [3 Marks]

Answer: 209

$Y = 4X^2 + 1 + 4X$

$E[Y] = 4E(X^2) + 1 + 4E(X).$

$Y = (2X+1)^2$ $\quad f_Y(y)$

$\vdots$

$P(X \geq c) \leq \frac{\mu}{c}$

3. Let X be a Poisson random variable with mean equal to 10. Which of the following bounds can be obtained using Markov's inequality? [3 marks]

$X \sim P(10)$

(a) $P(X > 20) \geq \frac{1}{2}$

(b) $P(X < 20) \leq \frac{1}{2}$

(c) $P(X > 25) \geq \frac{10}{25}$

(d) $P(X > 25) \leq \frac{10}{26}$

$P(X > 25) = P(X \geq 26) \leq \frac{10}{26}$

Answer: (d)

$P(X > 25) = P(X = 26) + P(X = 27) + \ldots$

$P(X \geq 25) = P(X = 25) + P(X = 26)$

4. Suppose X and Y are independent random variables with means 10 and 20, and variances 2 and 4, respectively. Find the value of $\text{Var}(XY)$. [3 Marks]

   Hint: If X and Y are independent, then X^2 and Y^2 are also independent.

   (a) 0
   (b) 40,000
   (c) 1208
   (d) Cannot be determined

   Answer : c

$$\int \frac{1}{x^2}\,dx = \int x^{-2}\,dx = \frac{x^{-2+1}}{-2+1}$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}$$

atleast 74

$X \geq 74$

$P(X \geq c) \leq \frac{\mu}{c}$

5. Suppose that a random variable X denotes the number of items produced in a factory during a week with mean 50 and variance 25.

   (i). Using Markov's inequality, find a bound on the probability that this week's production will exceed 74. [2 Marks]

   (a) $P(X \geq 74) \geq \frac{25}{37}$

   (b) $P(X \geq 75) > \frac{2}{3}$

   (c) $P(X \geq 75) \leq \frac{2}{3}$ ✓

   (d) $P(X \geq 74) < \frac{25}{37}$

   Answer : c

$P(X > 74) \leq \frac{50}{74}$

$P(X > 74) = P(X \geq 75) \leq \frac{50}{75}$

$\leq \frac{2}{3}$

   (ii). Using Chebyshev's inequality, find a lower bound on the probability that this week's production will be between 40 and 60. Enter the answer correct to two decimal places. [3 Marks]

   Answer : 0.75 ; Range : 0.72 to 0.78

$P(40 < X < 60) \geq 1 - \frac{1}{k^2}$

$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$

$P(40 - 50 < X - 50 < 60 - 50)$

$P(-10 < X - 50 < 10) = P(|X - 50| < 10) \geq 1 - \frac{1}{k^2}$

$k\sigma = 10$

$\Rightarrow k \cdot 5 = 10$

$\Rightarrow k = 2$

$\geq 1 - \frac{1}{4}$

$\geq \frac{3}{4}$

6. The CDF of a random variable X is given as

$$F_X(x) = \begin{cases} 1 - e^{-4x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

What is the value of $P(-4 < X \leq 6)$? [3 Marks]

(a) $e^{-20} - e^{-24}$

(b) $1 - e^{-24}$

(c) $e^{-24} - e^{-20}$

(d) e^{-24}

$P(a < X < b)$

$= F_X(b) - F_X(a)$

$P(-4 < X \leq 6) = F_X(6) - \underbrace{F_X(-4)}$

$= 1 - e^{-4 \cdot 6}$

$= 1 - e^{-24}$

Answer : b

7. Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that

$$f(x) = \begin{cases} e^{(1-ax)}, & 0 < x < \infty, \\ 0, & \text{otherwise}, \end{cases}$$

where a is any real constant. Find a such that the function f is a valid density function.
Hint: $\int e^{kx}\, dx = \frac{e^{kx}}{k}$ [3 marks]

$$\int_0^\infty f(x)\, dx = 1$$

(a) $\frac{1}{e}$
(b) $1 - e$
(c) $\frac{2}{e} - \frac{1}{e^2}$
(d) e

$$e \cdot e^{-ax}$$

$$\Rightarrow \int_0^\infty e^{(1-ax)}\, dx = 1$$

Answer: (d)

$$\Rightarrow e \int_0^\infty e^{-ax}\, dx = 1$$

$$\Rightarrow e \cdot \left. \frac{e^{-ax}}{-a} \right|_0^\infty = 1$$

$$\Rightarrow \frac{e}{a} - (0 - 1)$$

$$\Rightarrow \frac{e}{a}(-0 + 1) = 1$$

$$\Rightarrow \frac{e}{a} = 1$$

$$\Rightarrow a = e$$

8. Let X be a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} \dfrac{k}{(1+x)^2}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k. Enter the answer correct to two decimal places.

<u>Hint:</u> $\displaystyle\int \frac{1}{(a+bx)^2}dx = \frac{-1}{b(a+bx)}$ [3 Marks]

Answer: 1.25 ; Range: 1.23 to 1.27

$p(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$ $\quad p = \frac{1}{2}$

$= \binom{n}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x} = \binom{n}{x} \left(\frac{1}{2}\right)^n$

$$^nC_r = \frac{n!}{r!(n-r)!}$$

9. Suppose $X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$.

(i). Find the value of n for which $\frac{1}{30}P(X=3) = P(X=2)$. [3 Marks]

(a) 90
(b) 92
(c) 30
(d) 32

Answer : (b)

$$\frac{1}{30} \cdot \binom{n}{3}\cancel{\left(\frac{1}{2}\right)^n} = \binom{n}{2}\cancel{\left(\frac{1}{2}\right)^n}$$

$$\Rightarrow \frac{1}{30} \cdot \frac{\cancel{n!}}{3!(n-3)!} = \frac{\cancel{n!}}{2!(n-2)!}$$

$$\Rightarrow \frac{1}{30} \cdot \frac{1}{3 \times \cancel{2!}\,\cancel{(n-3)!}} = \frac{1}{\cancel{2!}\,(n-2)\cancel{(n-3)!}}$$

$$\Rightarrow \frac{1}{90} = \frac{1}{n-2}$$

$$\Rightarrow n-2 = 90$$

$$\Rightarrow n = 92$$

$P(|X-\mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$

(ii). Using the Chebyshev's inequality, find a lower bound for $P(-2\sigma \leq X - \mu \leq 2\sigma)$, where μ and σ^2 are mean and variance of X. Enter the answer correct to 2 decimal places [2 Marks]

Answer: 0.75

$P(-2\sigma \leq X - \mu \leq 2\sigma)$

$P(|X-\mu| \leq 2\sigma) \geq 1 - \frac{1}{4}$

$\geq \frac{3}{4}$

10. Sruthi throws a dart onto a circular board. Let a random variable X denote the distance from the center to the point where the dart hits the board. Suppose the PDF of X is

$$f_X(x) = \begin{cases} kx(1-x^2), & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Find the value of k. [2 Marks]

Answer : 4

$$\int_0^1 f(x)\,dx = 1$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\Rightarrow \quad k \int_0^1 (x - x^3)\,dx = 1$$

solve.

$$\Rightarrow k = 4$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

$$F_X(b) - F_X(a)$$

(ii) Find the value of $P\left(\frac{1}{4} \leq X \leq \frac{3}{4}\right)$. Enter the answer correct to two decimal places.

[3 Marks]

Answer : 0.69 ; Range : 0.66 to 0.72

$$\int_{1/4}^{3/4} 4(x - x^3)\,dx$$

$$= 0.69$$

11. Rajat and Meenal are playing a game. Rajat will roll a fair six-sided die, and Meenal will flip a fair coin as many times as the number shown on the die. Let X be the number displayed on the die, and Y be the number of heads obtained by Meenal.

(i). If $X = 4$, then which of the following options are true? [3 Marks]

(a) Range of $(Y|X = 4) = \{1, 2, 3, 4\}$

(b) Range of $(Y|X = 4) = \{0, 1, 2, 3, 4\}$

(c) $(Y|X = 4) \sim \text{Binomial}\left(4, \frac{1}{6}\right)$

(d) $(Y|X = 4) \sim \text{Binomial}\left(4, \frac{1}{2}\right)$

Answer : b, d

4

Roll die :→ 4

toss a coin 4 time

'0'

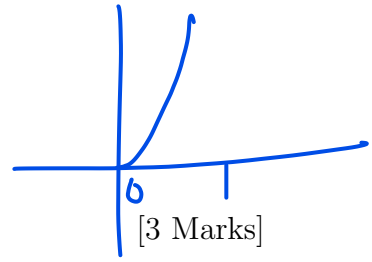
$(Y \mid X = 4) \sim Bin\left(4, \frac{1}{2}\right)$

(ii). Find the value of $100P(Y = 6)$. Enter the answer correct to two decimal places. [2 Marks]

Answer : 0.26 ; Range : 0.23 to 0.29

12. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 4x^3, & 0 < x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



Find the PDF of $Y = X^3$. [3 Marks]

Theorem.

(a) $f_Y(y) = \begin{cases} \frac{1}{3}\left(\frac{y}{4}\right)^{1/3}, & 0 < y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$

(b) $f_Y(y) = \begin{cases} \frac{4}{3}(y)^{1/3}, & 0 < y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$

(c) $f_Y(y) = \begin{cases} \frac{1}{3}(y)^{1/3}, & 0 < y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$

(d) $f_Y(y) = \begin{cases} 4(y)^{1/3}, & 0 < y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$

Answer : (b)

$E[XY] = E[X] \cdot E[Y]$

13. Suppose X and Y are two independent random variables with probability density functions as

$$g(x) = \begin{cases} \dfrac{8}{x^3}, & x > 2, \\ 0, & \text{otherwise.} \end{cases}$$

and

$$h(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the value of $E[XY]$. [3 Marks]

(a) 2

(b) $\frac{2}{3}$

(c) $\frac{8}{3}$

(d) Insufficient information

Answer : c

$E[X] = \int_2^\infty x \cdot \frac{8}{x^3}\, dx =$

$E[Y] = \int_0^1 y \cdot 2y\, dy =$